ERCOT Project Website Companion Report

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The project website brings our ERCOT demand forecasting work together in one organized place. It provides access to all four required deliverables and adds an interactive way to examine the evidence behind our conclusions. This companion report explains where the materials can be found, how the website works, and why developing this format became part of my goal as the data visualizer and presenter.

Project website: https://mjscarpello.github.io/DSC450-ERCOT-Presentation/

Project repository: https://github.com/mjscarpello/DSC450-ERCOT-Presentation

Required deliverables and direct links

Each link below opens the identified final material. The website is an additional way to present and explore the completed work.

Required deliverable	Where to find it
Report or white paper	Final Milestone 2 research report PDF https://mjscarpello.github.io/DSC450-ERCOT-Presentation/research/reports/Milestone_2_ERCOT_Demand_Prediction_FINAL.pdf The approved report is in research/reports.
Presentation	Watch the final presentation on the website https://mjscarpello.github.io/DSC450-ERCOT-Presentation/#presentation Presentation release and MP4 download https://github.com/mjscarpello/DSC450-ERCOT-Presentation/releases/tag/project2-milestone3-final The embedded player is on the homepage. The complete video runs 18 minutes 39 seconds.
Q and A	Ten stakeholder questions and answers https://github.com/mjscarpello/DSC450-ERCOT-Presentation/blob/main/research/qa/stakeholder_questions.md The written Q&A is in research/qa and corresponds to the ten questions in the final report.
Code or link to code	GitHub project repository https://github.com/mjscarpello/DSC450-ERCOT-Presentation Final visualization notebook with saved outputs https://github.com/mjscarpello/DSC450-ERCOT-Presentation/blob/main/research/notebooks/originals/DSC450_ERCOT_Full_Visualization_Hybrid_model.ipynb Notebook guide and reproduction instructions https://github.com/mjscarpello/DSC450-ERCOT-Presentation/blob/main/research/docs/NOTEBOOK_GUIDE.md The research folder contains notebooks, supplied data, analysis code and validation evidence.

The final research report remains the primary written analysis. This companion report explains the website and how the submission is organized.

Website features and their purpose

Open the interactive Research Explorer
https://mjscarpello.github.io/DSC450-ERCOT-Presentation/ERCOT%20project%20context%20web/explorer.html

Embedded presentation and final materials. Visitors can watch the completed video directly on the homepage using playback, seeking, volume and full-screen controls. Links below the player open the report, written Q&A and final notebook. A video download provides another way to access the presentation.

Interactive connection map. The explorer organizes 178 research concepts and 591 recorded relationships. Selecting a concept shows its connections to findings, figures, metrics, sources and limitations. The Follow control centers a related concept, allowing the reader to move through the reasoning rather than follow a fixed slide sequence.

Relationship explanations and filters. Why linked explains the recorded relationship and its supporting context. Filters focus the map on particular categories. Arrowheads show documented direction, while the legend distinguishes dependencies from associations. These visual links help explain the research without implying that every connection is causal.

Trace a path. Readers can choose two research records and inspect a route through their recorded connections. This makes it possible to explore how a research question relates to a model result or figure. A displayed path is an explanation route, not statistical or causal proof.

Project overview and research findings. The overview brings key comparison statistics together with interpretation cautions. Findings can be filtered by keyword and opened for more detail. This gives readers a starting point before they examine the supporting evidence.

Model comparison. The comparison view displays mean absolute error for eight forecast series and switches between overall evaluation and the highest one percent of observed demand. This makes the change in model performance visible and supports the distinction between average accuracy and performance during extreme hours.

Original figure library. The explorer includes 38 original notebook figures, grouped for browsing and searchable by topic, title or figure identifier. Opening a figure provides a larger view and research context. The separate final notebook preserves the approved visualization work and its saved outputs.

Search and research questions. Search helps readers locate research records without knowing where they appear in the map. The research questions section brings together report appendix questions and figure-related questions, keeping the purpose of the analysis connected to the evidence used to answer it.

Source registry and detailed records. Source records and detail views identify where statements, values and figures originated. Recorded status labels distinguish verified arithmetic from report statements, interpretations and open questions. A source locator identifies evidence; it does not guarantee that every historical source file is available.

Why I created this website

As the data visualizer and presenter, I wanted the final submission to show how our work fits together. Sarah, Andrea and I contributed different parts of the project, and I felt that presenting the results clearly also meant making the relationships between those contributions easier to understand. My goal became to give the audience a way to move from a conclusion to the figure, model comparison or source that supports it.

The recorded presentation provides the story in an order we chose. The website lets a reader return to a particular question and examine it at their own pace. For example, someone interested in the hybrid forecast can compare its overall error with its error during the highest-demand hours, then inspect related findings and limitations. That connection matters because a strong yearly average does not fully explain how a forecast behaves when demand is especially high.

I wanted the final result to feel like a culmination of the analysis and the communication work. Organizing the report, presentation, Q&A and code under clear links makes the submission easier to review. Connecting the research within the explorer gives those materials additional context and makes the work useful beyond a single viewing of the video.

How the website extends the assignment

The assignment requests a report or white paper, a presentation, Q&A, and code or a link to code. An interactive website is additional to those four stated deliverables. I chose to go beyond that requested format by creating a public project hub, an embedded presentation and an explorable record of the research connections. The required materials remain individually identified and directly accessible on the first page of this report and in the repository README.

The added value comes from what a reader can do with the work: locate a deliverable, inspect a figure, compare evaluation conditions and follow a claim back toward its source. This reflects the kind of presentation I wanted to create, where visual communication also supports careful review of how the conclusions were reached.

Research accountability and supporting resources

The repository preserves the approved final report and complete 104-cell final visualization notebook, alongside separate reproduction workflows, supplied CSVs, source hashes and validation results. Git history records revisions and helps distinguish final materials from earlier research versions. The website is a reference to the study, not a live forecasting service.

The scientific limitations remain visible. Historical observed weather is not the same as an advance weather forecast, the highest-demand subset is selected retrospectively, and matched target hours do not establish equal forecast issue times. The original raw data snapshot and failed static SARIMA prediction trace are unavailable. Those boundaries are part of presenting the project responsibly.

Notebook guide
https://github.com/mjscarpello/DSC450-ERCOT-Presentation/blob/main/research/docs/NOTEBOOK_GUIDE.md
Validation results
https://github.com/mjscarpello/DSC450-ERCOT-Presentation/blob/main/research/validation/REPOSITORY_CHECKS.json
Scientific and reproduction limitations
https://github.com/mjscarpello/DSC450-ERCOT-Presentation/blob/main/research/docs/LIMITATIONS.md